

Numerical Methods

Chapter-1

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Course Code: CSE 2203

Course Title : Numerical Methods

Text Book:

S.S. Sastry, “Introductory Methods of Numerical Analysis”, 5th Edition.

Reference Books:

Steven. C Chapra, “Numerical Methods for Engineers, McGraw-Hill

S.S. Kuo, “Computer Applications of Numerical Methods”, Addison-Wesley.

Numerical Methods:

Algorithms that are used to obtain numerical solutions of a mathematical problem.

Why do we need them?

1. No analytical solution exists,
2. An analytical solution is difficult to obtain or not practical.

Numerical Approximations:

Numerical methods is an area of study in mathematics that discusses the solutions to various mathematical problems involving differential equations, curve fittings, integrals, Eigen values and root findings through approximations rather than exact solutions.

Read thoroughly the following article from chapter-1:

Article-1.1, 1.1.1, 1.1.2, 1.1.3

Task-1: Why do we read numerical analysis course as a CSE student?

Task-2: Write some real life application of Numerical Analysis.

Errors and Their Computation

Two kinds of numbers:

1. Exact number: An exact number is one that has no uncertainty. Example- $1, 2, 3, \dots, 1/2, 3/2, \sqrt{2}, \pi, e$ etc.

2. Approximate number: An approximate number is one that does have uncertainty. Example approximate value of π is 3.1416, approximate value of $1/3$ is 0.333 etc.

Significant Digits: The digits that are used to express a number are called significant digits or significant figures. For example-

Scenario-1: The numbers 3.1416, 0.33333 and 1.73205 contain 5, 5 and 6 significant digits respectively.

Scenario-2: The numbers 0.0023, 0.000145 and 0.0000145 contain 2, 3 and 3 significant digits respectively. In this case zeros serve only to fix the position of the decimal point.

Rounding/Round off: In numerical computations, we come across numbers which have large number of digits and it will be necessary to cut them to a usable number of figures or digits. This process is called rounding off.

Rules of rounding off a number: To round off a number to n significant digits, discard all digits to the right of the n th digit and if this discarded number is-

1. Less than half a unit in the n th place, leave the n th digit unaltered.
2. Greater than half a unit in the n th place, increase the n th

Rules of rounding off a number (Cont'd)

digit by unity.

3. Exactly half a unit in the n th place, increase the n th digit by unity if it is odd, otherwise leave it unchanged.

The number thus rounded off is said to be correct to n significant figures.

Example- If the numbers 1.6583, 30.0567, 0.859378 and 3.14159 are rounded off to four significant digits then we will get 1.658, 30.06, 0.8594 and 3.142 respectively.

Absolute, Relative and Percentage Errors

Absolute error: It is the numerical difference between the true/exact value of a quantity and its approximate value.

$$E_A = X - X_1 = \delta X$$

Where, E_A = absolute error

X = true/exact value

X_1 = approximate value

Absolute, Relative and Percentage Errors (Cont'd)

Relative error: It is the ratio of absolute error and true/exact value of a quantity.

$$E_R = \frac{E_A}{X} = \frac{\delta X}{X}$$

Where, E_R = relative error

E_A = absolute error

X = true/exact value

Percentage error: It is the 100 times of relative error. It is given by-

$$E_P = 100 E_R$$

Absolute, Relative and Percentage Errors(Cont'd)

Absolute accuracy: If δX be a number then the quantity $|X_1 - X| \leq \delta X$ measures the absolute accuracy.

Relative accuracy: If δX be a number then the quantity $\frac{\delta X}{|X|} \approx \frac{\delta X}{|X_1|}$ measures the relative accuracy.

Percentage accuracy: If δX be a number then the quantity $\frac{\delta X}{X} \times 100$ measures the percentage accuracy.

Absolute, Relative and Percentage Errors(Cont'd)

Formula: If the number X is rounded to N decimal places, then-

$$\delta X = \frac{1}{2}(10^{-N})$$

Example: Find the percentage accuracy if $X=0.51$ and is correct to 2 decimal places.

Solution: Here, $X=0.51$

$$\begin{aligned}\delta X &= \frac{1}{2}(10^{-2}) \\ &= 0.005\end{aligned}$$

Absolute, Relative and Percentage Errors(Cont'd)

$$\begin{aligned}\text{So, percentage accuracy} &= \frac{\delta X}{X} \times 100 \% \\ &= \frac{0.005}{0.51} \times 100 \% \\ &= 0.98 \%\end{aligned}$$

Self study: Article-1.4: A General Error Formula

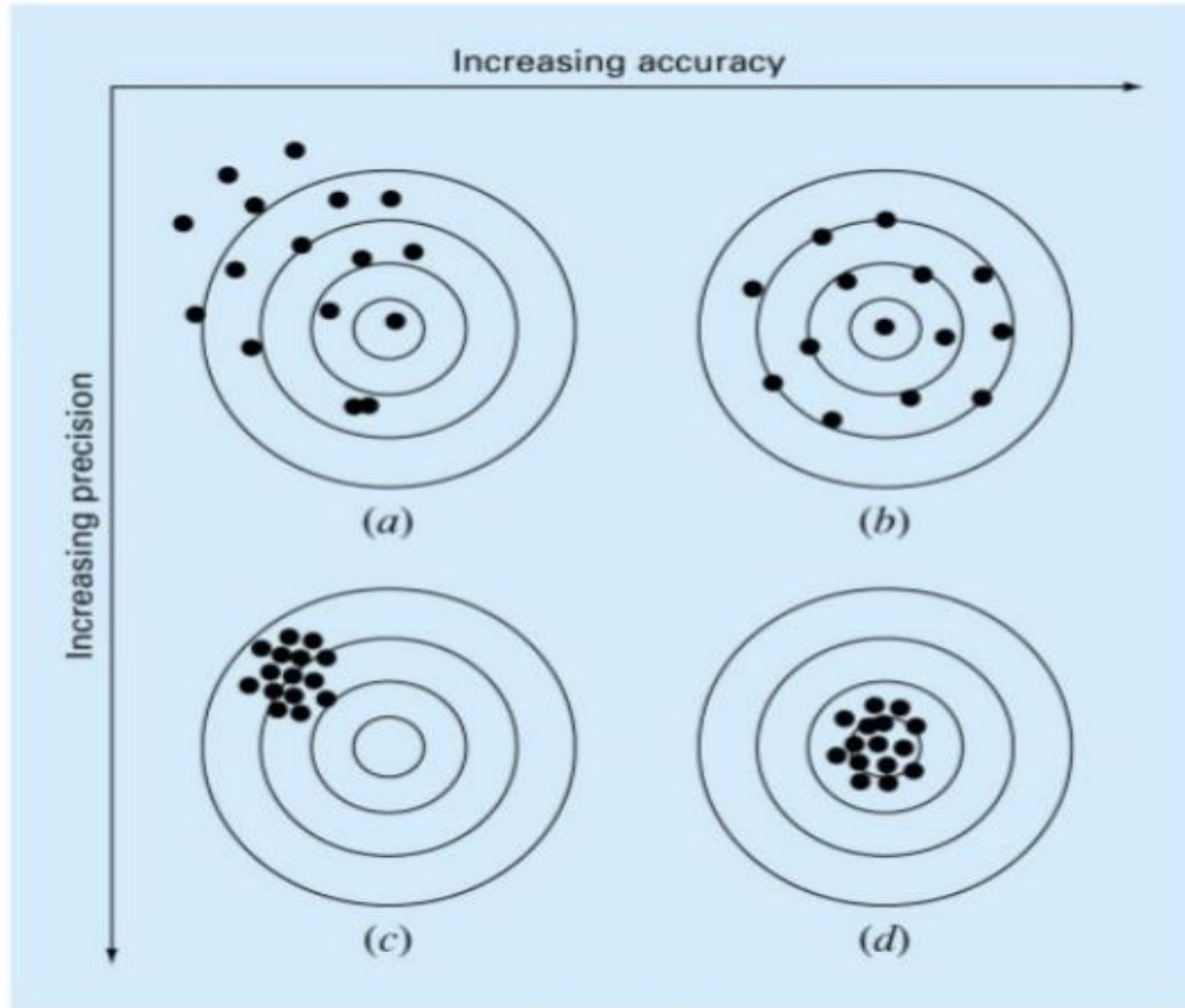
Example- 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.10, 1.11.

Accuracy and Precision

Accuracy: The accuracy of an approximate number is given by the number of significant digits in it. It measures about how close your solution is.

Precision: The precision of an approximate number is given by the position of the rightmost significant digit. It measures about how close your repetition of the solution are.

Accuracy and Precision (Cont'd)



Task:

Do the Exercises- 1.1, 1.2, 1.3, 1.5, 1.6

Any Question???